

20260401A Number Crunch

Background and Problem Intuition

Mathematical process: Start from Bayes' rule, then convert inference into optimisation.

$$p(\boldsymbol{\theta} \mid \mathbf{y}) = \frac{p(\mathbf{y} \mid \boldsymbol{\theta})p(\boldsymbol{\theta})}{p(\mathbf{y})}, \quad (1)$$

$$p(\mathbf{y}) = \int p(\mathbf{y}, \boldsymbol{\theta}) d\boldsymbol{\theta}, \quad (2)$$

where $p(\mathbf{y})$ is typically intractable.

Introduce a tractable variational density $q(\boldsymbol{\theta})$ and write

$$\log p(\mathbf{y}) = \log p(\mathbf{y}) \int q(\boldsymbol{\theta}) d\boldsymbol{\theta} \quad (3)$$

$$= \int q(\boldsymbol{\theta}) \log p(\mathbf{y}) d\boldsymbol{\theta} \quad (4)$$

$$= \int q(\boldsymbol{\theta}) f(\boldsymbol{\theta}) d\boldsymbol{\theta}, \quad f(\boldsymbol{\theta}) \equiv \log p(\mathbf{y}) \quad (5)$$

$$= \mathbb{E}_q[f(\boldsymbol{\theta})] \quad (6)$$

$$= \mathbb{E}_q[\log p(\mathbf{y})] \quad (7)$$

$$= \mathbb{E}_q \left[\log \frac{p(\mathbf{y}, \boldsymbol{\theta})}{p(\boldsymbol{\theta} \mid \mathbf{y})} \right] \quad (8)$$

$$= \mathbb{E}_q \left[\log \left(\frac{p(\mathbf{y}, \boldsymbol{\theta})}{q(\boldsymbol{\theta})} \cdot \frac{q(\boldsymbol{\theta})}{p(\boldsymbol{\theta} \mid \mathbf{y})} \right) \right] \quad (9)$$

$$= \mathbb{E}_q \left[\log \frac{p(\mathbf{y}, \boldsymbol{\theta})}{q(\boldsymbol{\theta})} \right] + \mathbb{E}_q \left[\log \frac{q(\boldsymbol{\theta})}{p(\boldsymbol{\theta} \mid \mathbf{y})} \right] \quad (10)$$

$$= \mathbb{E}_q[\log p(\mathbf{y}, \boldsymbol{\theta})] - \mathbb{E}_q[\log q(\boldsymbol{\theta})] + \mathbb{E}_q \left[\log \frac{q(\boldsymbol{\theta})}{p(\boldsymbol{\theta} \mid \mathbf{y})} \right]. \quad (11)$$

Sequence of events (plain-language):

1. Start from $\log p(\mathbf{y})$, which is constant with respect to $\boldsymbol{\theta}$.
2. Multiply by 1 using normalisation of q : $\int q(\boldsymbol{\theta}) d\boldsymbol{\theta} = 1$.
3. Rewrite as an integral: $\log p(\mathbf{y}) = \int q(\boldsymbol{\theta}) \log p(\mathbf{y}) d\boldsymbol{\theta}$.
4. Recognise expectation form: $\int q(\boldsymbol{\theta}) f(\boldsymbol{\theta}) d\boldsymbol{\theta} = \mathbb{E}_q[f(\boldsymbol{\theta})]$.
5. Use Bayes identity inside the logarithm: $p(\mathbf{y}) = \frac{p(\mathbf{y}, \boldsymbol{\theta})}{p(\boldsymbol{\theta} \mid \mathbf{y})}$.

6. Insert $q(\boldsymbol{\theta})/q(\boldsymbol{\theta})$ to create ELBO and KL terms.
7. Split product and quotient with log rules, then split expectations linearly.
8. Identify

$$\text{ELBO}(q) = \mathbb{E}_q[\log p(\mathbf{y}, \boldsymbol{\theta})] - \mathbb{E}_q[\log q(\boldsymbol{\theta})]$$

and

$$\text{KL}(q(\boldsymbol{\theta}) \parallel p(\boldsymbol{\theta} \mid \mathbf{y})) = \mathbb{E}_q \left[\log \frac{q(\boldsymbol{\theta})}{p(\boldsymbol{\theta} \mid \mathbf{y})} \right].$$

Define

$$\text{ELBO}(q) = \mathbb{E}_q[\log p(\mathbf{y}, \boldsymbol{\theta})] - \mathbb{E}_q[\log q(\boldsymbol{\theta})] \tag{12}$$

$$= \mathbb{E}_q[\log p(\mathbf{y}, \boldsymbol{\theta})] + H(q), \tag{13}$$

so that

$$\log p(\mathbf{y}) = \text{ELBO}(q) + \text{KL}(q(\boldsymbol{\theta}) \parallel p(\boldsymbol{\theta} \mid \mathbf{y})). \tag{14}$$

Since $\text{KL}(\cdot \parallel \cdot) \geq 0$, maximising $\text{ELBO}(q)$ is equivalent to minimising divergence to the true posterior.

Next Steps

Alignment with Course Requirements

Timeline

Summary